Homework 04

INT3405

October 2, 2024

Collaboration: List the names of all people you have collaborated with and for which question(s). You may discuss problems with other students, but you must not view or copy another student's written answers. Each student is responsible for independently understanding, writing, and submitting their own work.

Acknowledgments: If you reference solutions from published sources, the web, or other textbooks, you must acknowledge these sources. It's acceptable to use existing proofs for guidance, but this must be noted, and you should first attempt to solve the problem on your own. All students are expected to fully comprehend the steps they include in their written work.

Exercise 1. Given a dataset with two classes, where the feature matrix X and the corresponding label vector y are:

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \quad y = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

Use the normal equation to compute the weight vector w for the classification problem, where the hypothesis function is:

$$h(X) = Xw$$

Exercise 2. You are developing a system using **Naive Bayes** to determine whether the weather is suitable for playing badminton. You are given a dataset of 10 days of weather data and whether badminton was played on those days. The features include *Outlook* (sunny, rainy, overcast), *Temperature* (high, medium, low), *Humidity* (high, low), and *Wind* (yes or

no). The dataset is summarized in the following table:

Day	**Outlook**	**Temperature**	**Humidity**	**Wind**	**Play Badminton**
1	*Sunny*	*High*	*High*	*No*	*No*
2	*Sunny*	*High*	*High*	*Yes*	*No*
3	*Overcast*	*High*	*High*	*No*	*Yes*
4	*Rainy*	*Medium*	*High*	*No*	*Yes*
5	*Rainy*	*Low*	*Low*	*No*	*Yes*
6	*Rainy*	*Low*	*Low*	*Yes*	*No*
7	*Overcast*	*Low*	*Low*	*Yes*	*Yes*
8	*Sunny*	*Medium*	*High*	*No*	*No*
9	*Sunny*	*Low*	*Low*	*No*	*Yes*
10	*Rainy*	*Medium*	*High*	*Yes*	*Yes*

A new day is observed with the following weather conditions: - **Outlook**: Overcast - **Temperature**: Medium - **Humidity**: High - **Wind**: No. Using Naive Bayes Classification and Maximum-Likelihood Estimation (MLE) to determine whether this day is suitable for playing badminton.

Exercise 3. You are a data analyst working with a dataset to predict whether a student will pass an exam based on the number of hours they study. The dataset includes 10 students with their study hours and exam results as follows:

Student	**Study Hours**	**Pass Exam (1: Yes, 0: No)**
1	1	0
2	2	0
3	3	0
4	4	1
5	5	1
6	6	1
7	7	1
8	8	1
9	9	1
10	10	1

Question: Using logistic regression, predict the probability that a student will pass the exam if they study for 5 hours.